Project 3: Deep Hedging

(First discussion: Nov 4; Last questions: Nov 18; Deadline: Nov 25; In charge Arishi Orra)

The goal of this project is to implement the deep hedging method of Buehler et al. (2019) from scratch[1] and test it on data simulated from a Black–Scholes and a Heston model. Submit your solution using the template `template3.ipynb`. One submission per team is enough.

1. First, consider the Black–Scholes model, which describes the evolution of the stock price S with an SDE of the form
$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 = s_0 \in \mathbb{R}_+, \tag{1}$$
where W is a Brownian motion under a risk-neutral probability measure $\mathbb{Q}$, σ is the volatility and r the risk-free interest rate, which we assume to be zero[2].

   Given an option with payoff $g(S_T)$ at maturity T, we aim to find a hedging strategy, i.e. a trading strategy that replicates the option payoff. It can be written as a solution[3] to the following optimization problem:
$$\inf_{H \in \mathcal{H}} \mathbb{E}\left[\left(p + \int_0^T H_u dS_u - g(S_T)\right)^2\right],$$
where $p = \mathbb{E}[g(S_T)]$ is the fair option price and $\mathcal{H}$ the set of all predictable processes.

   We can solve this problem numerically on a uniform time grid $0 = t_0 < t_1 < \ldots < t_N = T$ by approximating the Itô integral with a discrete stochastic integral $\sum_{j=0}^{N-1} H_j \cdot \left(S_{t_{j+1}} - S_{t_j}\right)$ for N neural networks $H_0, \ldots, H_{N-1}$ jointly trained by minimizing the empirical loss
$$\frac{1}{m}\sum_{i=1}^{m}\left(p + \sum_{j=0}^{N-1} H_j \cdot \left(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}\right) - g\left(s_T^{(i)}\right)\right)^2 \tag{2}$$
on a training set $D = \left\{(s_{t_0}^{(i)}, s_{t_1}^{(i)}, \ldots, s_{t_N}^{(i)}),\ 0 \le i \le m\right\}$ of m simulated paths of S.

   Implement and test the model following the steps below:

   a) Use Itô's formula to check that $S_t = s_0 \exp\left(\sigma W_t - \frac{1}{2}\sigma^2 t\right)$ is a solution of the SDE (1) (it is called *geometric Brownian motion*).

   b) Simulate a training set of 100,000 paths and a test set of 10,000 paths of the stock price S with parameters $N = 30$, $S_{t_0} = s_0 = 1$, $T = 3$ months $= 1/4$ and $\sigma = 0.35$.

[1] You can implement your model either in PyTorch (by adapting the code in `demo.ipynb`) or TensorFlow. In either case your model must be implemented from scratch: you are not allowed to use third-party repositories with ready-made implementations of deep hedging.

[2] This assumption is without loss of generality. The case $r \neq 0$ can always be reduced to the case $r = 0$ by working with discounted payoffs.

[3] The existence of a solution is guaranteed by the fact that the Black-Scholes model is complete.

The process S can be simulated exactly on a finite grid by setting

$$S_{t_{j+1}} = S_{t_j} \exp\left(-\frac{\sigma^2}{2}\frac{T}{N} + \sigma\sqrt{\frac{T}{N}} Z_{j+1}\right),$$

where $Z_1, \dots, Z_N$ are N iid standard Gaussian random variables.

c) Implement the model by specifying each $H_j \colon \mathbb{R} \to \mathbb{R}$ as a neural network with input S_{t_j}.

d) Train the hedging strategy for a call option with payoff $g(S_T) := (S_T - K)^+$ with strike $K = 1$ and maturity $T = 3$ months $= 1/4$ by minimizing the loss (2) on the training set.

Recall that in the Black–Scholes model, the fair price of a European call option at time t is given by

$$C(t, S_t) = \Phi(d_+) S_t - \Phi(d_-) K e^{-r(T-t)}, \tag{3}$$

where

$$d_+ = \frac{1}{\sigma\sqrt{T-t}}\left(\log\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)\right), \quad d_- = d_+ - \sigma\sqrt{T-t}$$

and Φ is the standard normal cumulative distribution function.

You can compute the time-0 price $p = C(0, S_0)$ using the Black–Scholes formula (3).

e) Evaluate the P&L of the hedged position $p + \sum_{j=0}^{N-1} H_j \cdot \left(S_{t_{j+1}} - S_{t_j}\right) - g\left(S_T\right)$ on the test set. Plot the histogram and print the empirical mean and standard deviation.

f) In the Black–Scholes model, the hedging problem admits an analytical solution given by

$$H_t^{\mathrm{BS}}(s) = \frac{\partial C(t,s)}{\partial s}.$$

i) Derive a closed-form formula for $H_t^{\mathrm{BS}}(s)$ by computing the partial derivative of (3).

ii) Evaluate the hedging losses on the test set when using the analytical hedging strategy H^{BS} to rebalance the hedging strategy at the trading times $t_0, t_1, \dots, t_{N-1}$. Plot their histogram and print their empirical mean and standard deviation. Compare with the results obtained in (e) above.

g) For $j \in \{0, 10, 20, 29\}$, compare the neural network function $s \mapsto H_j(s)$ with the analytical solution $s \mapsto H_{t_j}^{\mathrm{BS}}(s)$ in a plot for $s \in [0.5, 1.5]$. For which times t_j are the two functions most similar? Why?

2. Now, consider the Heston model, which models the risk-neutral dynamics of the stock price as

$$\begin{cases} dS_t &= \sqrt{V_t} S_t dW_t, \quad S_0 = s_0 \in \mathbb{R}_+, \\ dV_t &= a(b - V_t)dt + \sigma\sqrt{V_t} dW_t', \quad V_0 = v_0 \in \mathbb{R}_+ \end{cases} \tag{4}$$

for two Brownian motions W and W' with instantaneous correlation $\rho \in [-1, 1]$.

The Heston model is incomplete, which means that not every option payoff can be replicated by trading in the underlying stock. As an alternative, one can look for a hedging strategy that minimizes

$$p = \inf_{H \in \mathcal{H}} R\left(g(S_T) - \int_0^T H_u dS_u\right), \tag{5}$$

for a risk measure R. p denotes the minimum amount of money needed to compensate for the hedging risk under the risk measure R and is therefore the price that we would charge for selling the option.

As risk measure R one can, for instance, use Average Value at Risk at level $\alpha \in (0,1)$, given by

$$\mathrm{AVaR}_\alpha(X) = \frac{1}{1-\alpha}\int_\alpha^1 q_u(X)du,$$

where $q_u(X)$ is the u-quantile of X. It is well known that AVaR_α admits the variational representation

$$\mathrm{AVaR}_\alpha(X) = \inf_{w\in\mathbb{R}}\left(w + \frac{\mathbb{E}\left[(X-w)^+\right]}{1-\alpha}\right);$$

see e.g. Föllmer and Schied (2004). Therefore, (5) can be written as an empirical loss minimization problem with loss

$$\frac{1}{m}\sum_{i=1}^m \left(w + \frac{1}{1-\alpha}\left(g(s_T^{(i)}) - \sum_{j=0}^{N-1} H_j\cdot\left(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}\right) - w\right)^+\right), \tag{6}$$

where one simultaneously minimizes over $w\in\mathbb{R}$ and the N neural networks $H_0,\dots,H_{N-1}$.

Implement and test the approach following the steps below:

a) Simulate a training set of 100,000 paths and a test set of 10,000 paths of the stock price S and the variance process V with parameters $N=30$, $s_0=1$, $T=3$ month $=1/4$, $\rho=-0.3, a=4, b=0.5, v_0=0.5$ and $\sigma=1$.

   i) First simulate V. The transition density of V in (4) is known explicitly (Glasserman, 2004, Chapter 3.4) and can be simulated exactly on a finite grid by setting $V_{t_{j+1}} = c\cdot C_j$, where $c=\dfrac{\sigma^2}{4a}(1-e^{-aT/N})$ and each C_j is a non-central chi-square random variable with $\dfrac{4ab}{\sigma^2}$ degrees of freedom and $e^{-aT/N}(V_{t_j}/c)$ non-centrality parameter.
   (The non-central chi-square distribution is implemented in `scipy.stats.ncx2`*.)*

   ii) Now simulate S using the simplified Broadie–Kaya scheme (Andersen et al., 2010):

$$S_{t_{j+1}} = S_{t_j}\exp\left(\frac{\rho}{\sigma}\left(\left(V_{t_{j+1}}-V_{t_j}\right) - ab\frac{T}{N}\right) + \left(\frac{a\rho}{\sigma}-\frac{1}{2}\right)V_{t_j}\frac{T}{N} + \sqrt{(1-\rho^2)V_{t_j}\frac{T}{N}}\,Z_{j+1}\right)$$

   where $Z_1,\dots,Z_N$ are iid standard Gaussian random variables.

b) Implement the approach by specifying each H_j as a neural network with input (S_{t_j}, V_{t_j}).

c) Train the model for the same European call option as in 1.d) above by minimizing the loss (6) on the training set for two different AVaR levels, $\alpha=0.5$ and $\alpha=0.9$.

d) Compare the prices p for both values of α by evaluating the loss (6) on the test set for each trained model. Which is one higher? Why?

e) Compare the P&L of the hedged position $p+\sum_{j=0}^{N-1} H_{t_j}\left(S_{t_{j+1}}-S_{t_j}\right) - g\left(S_T\right)$ on the test set for both values of α. Plot the two histograms and report the empirical means and standard deviations.

References

Andersen, L. B., Jäckel, P., and Kahl, C. (2010). Simulation of square-root processes. *Encyclopedia of Quantitative Finance*, pages 1642–1649.

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Föllmer, H. and Schied, A. (2004). *Stochastic Finance: An Introduction in Discrete Time*, volume 3. De Gruyter.

Glasserman, P. (2004). *Monte Carlo Methods in Financial Engineering*, volume 53. Springer.